

Vishnu Koraganji

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Professional Summary

MS in Computer Science from Penn State University specializing in AI/ML, NLP, Generative AI, and Agentic systems. Full Stack Engineer at ClearChartAI building a production, HIPAA-compliant medical AI platform — RAG pipelines, Vertex AI vector search, LLM integration, and Zero-Trust security on Google Cloud. Skilled at designing, shipping, and optimizing scalable, reliable ML systems using Python, PyTorch, and cloud-native architectures.

Technical Skills

Languages: Python, Java, C, JavaScript, SQL, React, R
Machine Learning & Deep Learning: (PyTorch, TensorFlow/Keras, Pandas, Scikit-learn, NLTK, MLFlow, DVC); Classification, Clustering, Regression, Recommender Systems, Information Retrieval, NLP, Statistics, Computer Vision, Time Series Forecasting, CNN, RNN, YOLO, Ranking and Searching
Gen AI Tech Stack: Transformer models-BERT, GPT, Gemini, Llama, RAG, PEFT (LoRA & QLoRA) - Fine Tuning LLM, Langchain, CrewAI, Hugging Face, Sentence Transformers, Vector Search, FAISS, FastAPI
MLOps: Building scalable, reproducible ML pipelines; experiment tracking (MLFlow), model/data versioning (DVC), Docker containers, CI/CD/CT for machine learning.
Databases & Cloud & Tools: ChromaDB, PostgreSQL, Firestore, Firebase, Ollama, Ngrok, AWS, GCP (Vertex AI, Cloud Run, Cloud Functions), ArcGIS/QGIS, Git, VS Code
Relevant Coursework: AI/ML, NLP, Statistics, DSA & OS, Algorithms Analysis, DBMS & Data Analytics

Experience

Full Stack Engineer | ClearChartAI, Remote Aug 2025 - Present

- Architected and deployed a **HIPAA-compliant** medical AI platform end-to-end (React/Vite front end, **FastAPI** on Cloud Run, **Gen2 Cloud Functions**) on Google Cloud, serving source-cited **RAG** Q&A over patients' longitudinal health records with **sub-2s** streaming responses.
- Built the document-ingestion and retrieval pipeline (**Gemini Vision** extraction, semantic chunking, **Vertex AI** vector search over **768-dim** embeddings) and led a **zero-downtime** migration of **3** microservices to the google-genai SDK, re-indexing **5k** vectors and fixing **5** silent data-loss bugs.
- Implemented a **7-layer Zero-Trust** security model — MFA/**AAL2** (TOTP), JWT validation, and HIPAA audit logging across all PHI endpoints — meeting **NIST 800-63B** requirements.
- Engineered **CommonWell/TEFCA** health-information-exchange integration (**mTLS**, RS256 JWTs, **FHIR R4**) with **IAL2** identity verification, enabling patient-directed record retrieval across the U.S. national health network.

Graduate Teaching Assistant | Penn State University, Middletown, PA Aug 2023 - May 2025

- Assessed **10+** final-year CS projects with detailed written feedback and guided teams on Agile/SDLC best practices in Software Engineering (UML, use-case design).
- Facilitated learning for Data Structures & Algorithms and Calculus by reviewing tests/exams, emphasizing algorithmic complexity and optimization relevant to large-scale systems.

Cloud Architect Intern | AWS (Amazon Web Services), Vijayawada, IND Oct 2021 - Dec 2021

- Provisioned and managed core AWS services (**EC2**, **S3**, **RDS**, **VPC**) to build reliable cloud infrastructure with auto-scaling and load balancing.
- Secured environments through IAM access controls, encryption, and CloudWatch monitoring.

Projects

AI Agentic Resume Tailor (CrewAI, FastAPI, React, LangChain, Gemini API, Scrapy) Mar 2025 - May 2025

- Developed a CrewAI multi-agent system to automate resume tailoring, boosting ATS scores by **25-40%** via keyword optimization.
- Engineered a web-scraping agent parsing **5+** job boards (**95%** accuracy) and a PDF/DOCX/LaTeX pipeline with **100% format preservation**, deployed via FastAPI.

RAG Based QA Bot (PyTorch, DistilBERT, FAISS, FastAPI, LangChain, Ollama, Ngrok) Jan 2025 - Apr 2025

- Engineered a RAG chatbot for nuanced Q&A over the Amazon Reviews 2023 dataset (**213,593** products), integrating fine-tuned DistilBERT (**89%** validation accuracy) and FAISS + MMR retrieval.
- Achieved **95%** off-topic rejection and **0.85** cosine similarity, delivering sentiment-aware responses in **2-3s** via a FastAPI front-end.

Skin Cancer Detection System (Python, OpenCV, TensorFlow, PyTorch, YOLOv8, NumPy) Aug 2024 - Dec 2024

- Built a skin-cancer detection system using **YOLOv8** with a novel **SCSA attention module** and a preprocessing pipeline (artifact removal, CLAHE, color normalization).
- Evaluated on the HAM10000 dataset, achieving a **4% mAP@50** gain and **+8%** precision/recall over baseline YOLOv8.

Education

Penn State University Aug 2023 - May 2025

Masters in Computer Science Middletown, PA

VR Siddhartha Engineering College Aug 2019 - May 2023

Bachelors in Computer Science Engineering Vijayawada, India